

Mathematics for Computing

Module 2: Linear Algebra

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Topics Covered

1. Vectors
2. Coordinate System
3. Vector Addition
4. Vector Multiplication
5. Linear Combinations
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7. Matrix Multiplication as Composition
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15. Cross Products and Linear Transformations
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Vectors

Definition

A vector is a mathematical object having:

- ▶ Magnitude
- ▶ Direction

A vector in $\mathbb{R}^2$ is written as

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

Example

Consider

$$\mathbf{v} = \begin{bmatrix} 2 \\ -3 \end{bmatrix}.$$

Explanation: The vector moves 2 units in the x -direction and -3 units in the y -direction.

Vectors in AI and Machine Learning

A data point can be represented using a feature vector. For example, a student can be represented by

$$\mathbf{x} = \begin{bmatrix} \text{Study Hours} \\ \text{Attendance} \\ \text{Assignment Score} \end{bmatrix}.$$

Suppose

$$\mathbf{x} = \begin{bmatrix} 5 \\ 80 \\ 75 \end{bmatrix}.$$

Explanation: Each component represents one feature of the student.

- ▶ 5 = study hours
- ▶ 80 = attendance percentage
- ▶ 75 = assignment score

Thus, a student becomes a point in a multidimensional space.

Coordinate System

In a two-dimensional coordinate system,

$$\mathbf{v} = \begin{bmatrix} x \\ y \end{bmatrix}$$

represents the point (x, y) . **Example** Let

$$P = (-2, 4).$$

Then the position vector is

$$\overrightarrow{OP} = \begin{bmatrix} -2 \\ 4 \end{bmatrix}.$$

Explanation: Starting from the origin:

- ▶ Move 2 units left.
- ▶ Move 4 units upward.

Magnitude of a Vector

For

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix},$$

the magnitude is

$$\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2}.$$

New Example Let

$$\mathbf{v} = \begin{bmatrix} 5 \\ 12 \end{bmatrix}.$$

Then

$$\begin{aligned} \|\mathbf{v}\| &= \sqrt{5^2 + 12^2} \\ &= \sqrt{25 + 144} = \sqrt{169} = 13. \end{aligned}$$

Explanation: The vector has length 13 units.

Vector Addition

Vectors are added component by component. If

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix},$$

then

$$\mathbf{u} + \mathbf{v} = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \end{bmatrix}.$$

Example Let

$$\mathbf{u} = \begin{bmatrix} 1 \\ 4 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 3 \\ -2 \end{bmatrix}.$$

Therefore,

$$\mathbf{u} + \mathbf{v} = \begin{bmatrix} 1 + 3 \\ 4 - 2 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}.$$

Explanation: Simply add corresponding components.

Scalar Multiplication

Multiplying a vector by a scalar changes its magnitude.

$$c\mathbf{v} = \begin{bmatrix} cv_1 \\ cv_2 \end{bmatrix}.$$

Example

Let

$$\mathbf{v} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}.$$

For $c = -3$,

$$-3\mathbf{v} = \begin{bmatrix} -6 \\ 3 \end{bmatrix}.$$

Explanation:

Since the scalar is negative, the direction is reversed.

Properties of Vector Multiplication

For vectors $\mathbf{u}, \mathbf{v}$ and scalars a, b :

$$a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$$

$$(a + b)\mathbf{u} = a\mathbf{u} + b\mathbf{u}$$

$$(ab)\mathbf{u} = a(b\mathbf{u})$$

$$1\mathbf{u} = \mathbf{u}.$$

Simple idea:

Scalar multiplication follows the same basic algebraic rules used with ordinary numbers.

Linear Combination

A linear combination of vectors is

$$a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \cdots + a_k\mathbf{v}_k.$$

Example Let

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ -1 \end{bmatrix}.$$

Find

$$\begin{aligned} & 2\mathbf{v}_1 + 3\mathbf{v}_2. \\ &= 2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} + 3 \begin{bmatrix} 2 \\ -1 \end{bmatrix} \\ &= \begin{bmatrix} 2 \\ 4 \end{bmatrix} + \begin{bmatrix} 6 \\ -3 \end{bmatrix} = \begin{bmatrix} 8 \\ 1 \end{bmatrix}. \end{aligned}$$

Explanation: The resulting vector is constructed from the given vectors.

Span

The span of a set of vectors is the set of all possible linear combinations of those vectors.

$$\text{span}\{\mathbf{v}_1, \mathbf{v}_2\} = \{a\mathbf{v}_1 + b\mathbf{v}_2 : a, b \in \mathbb{R}\}.$$

Example

Let

$$\mathbf{v} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}.$$

Then

$$\text{span}\{\mathbf{v}\} = \left\{ a \begin{bmatrix} 1 \\ 2 \end{bmatrix} : a \in \mathbb{R} \right\}.$$

This represents a line passing through the origin.

Explanation: Changing a produces different points on the same line.

Basis

A basis is a set of vectors that:

1. Spans the vector space.
2. Is linearly independent.

Example

Consider

$$B = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}.$$

These vectors form a basis for $\mathbb{R}^2$.

For example,

$$\begin{bmatrix} 4 \\ 2 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 1 \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Explanation: Every vector in $\mathbb{R}^2$ can be represented using this basis.

Linear Independence

Vectors are linearly independent if

$$a_1 \mathbf{v}_1 + \cdots + a_k \mathbf{v}_k = \mathbf{0}$$

has only the solution

$$a_1 = a_2 = \cdots = a_k = 0.$$

Example

Consider

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} -2 \\ 1 \end{bmatrix}.$$

Neither vector is a scalar multiple of the other.

Therefore, they are linearly independent.

Matrix Multiplication as Composition

Suppose

$$T_1(\mathbf{x}) = A\mathbf{x}$$

and

$$T_2(\mathbf{x}) = B\mathbf{x}.$$

Applying T_1 first and then T_2 gives

$$T_2(T_1(\mathbf{x})) = B(A\mathbf{x}) = (BA)\mathbf{x}.$$

Therefore,

$$T_2 \circ T_1 \longleftrightarrow BA.$$

Explanation: Matrix multiplication represents applying one transformation after another.

Example: Composition of Transformations

Let

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

and

$$B = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}.$$

Then

$$\begin{aligned} BA &= \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 2 & 2 \\ 0 & 1 \end{bmatrix}. \end{aligned}$$

Explanation: The first matrix performs a shear, and the second stretches the x -direction. The product represents both transformations together.

Three-Dimensional Linear Transformation

A transformation

$$T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

can be represented by a 3×3 matrix. Consider

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix}.$$

For

$$\mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix},$$

we obtain

$$A\mathbf{x} = \begin{bmatrix} 2 \\ 6 \\ 4 \end{bmatrix}.$$

Explanation: The transformation stretches:

x-direction by 2, y-direction by 3, z-direction by 4.

3D Rotation About the z-Axis

A rotation by θ about the z-axis is

$$R_z(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Example

For $\theta = 180^\circ$,

$$R_z(180^\circ) = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Therefore,

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \rightarrow \begin{bmatrix} -1 \\ -2 \\ 3 \end{bmatrix}.$$

Explanation: The point rotates halfway around the z-axis.

Determinant

For

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

the determinant is

$$\det(A) = ad - bc.$$

It tells us about:

- ▶ Area scaling
- ▶ Orientation
- ▶ Invertibility

Example Let

$$A = \begin{bmatrix} 3 & 1 \\ 2 & 2 \end{bmatrix}.$$

Then

$$\det(A) = 3(2) - 1(2) = 6 - 2 = 4.$$

Determinant and Geometry

Consider

$$A = \begin{bmatrix} 3 & 0 \\ 0 & 2 \end{bmatrix}.$$

Then

$$\det(A) = 3(2) = 6.$$

Therefore, the area is multiplied by 6. **Example:**

If the original area is

$$A_0 = 2,$$

then the transformed area is

$$A_1 = 6(2) = 12.$$

Explanation: The determinant tells us how much a transformation changes area.

Inverse Matrices

The inverse of A is denoted by A^{-1} .

It satisfies

$$AA^{-1} = A^{-1}A = I.$$

For

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

This exists only when

$$ad - bc \neq 0.$$

Example: Inverse Matrix

Let

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}.$$

First,

$$\det(A) = 2(1) - 1(1) = 1.$$

Therefore,

$$A^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}.$$

Check:

$$AA^{-1} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Explanation: The inverse reverses the transformation represented by A .

Column Space

The column space of a matrix is the span of its columns.
Consider

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}.$$

The columns are

$$\mathbf{c}_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{c}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}.$$

Therefore,

$$\text{Col}(A) = \text{span} \{ \mathbf{c}_1, \mathbf{c}_2 \}.$$

Explanation: The column space contains every possible output $A\mathbf{x}$.

Null Space

The null space contains all vectors satisfying

$$A\mathbf{x} = 0.$$

Consider

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

Let

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}.$$

Then

$$A\mathbf{x} = \begin{bmatrix} x + y \\ y + z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Thus,

$$x + y = 0, \quad y + z = 0.$$

Taking $y = t$,

Nonsquare Matrices as Transformations

A matrix does not have to be square.

If

$$A \in \mathbb{R}^{m \times n},$$

then

$$A : \mathbb{R}^n \rightarrow \mathbb{R}^m.$$

Example

Consider

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

This is a 2×3 matrix.

Therefore,

$$A : \mathbb{R}^3 \rightarrow \mathbb{R}^2.$$

Nonsquare Matrices and Dimension Reduction

Consider

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Let

$$\mathbf{x} = \begin{bmatrix} 4 \\ 7 \\ 9 \end{bmatrix}.$$

Then

$$A\mathbf{x} = \begin{bmatrix} 4 \\ 9 \end{bmatrix}.$$

The second component is removed. **Explanation:** A transformation

$$\mathbb{R}^3 \rightarrow \mathbb{R}^2$$

can be used to reduce the number of dimensions.

Dot Product

For

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix},$$

the dot product is

$$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2.$$

Example Let

$$\mathbf{u} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}.$$

Then

$$\mathbf{u} \cdot \mathbf{v} = 2(1) + 1(-2) = 0.$$

Therefore,

$$\mathbf{u} \perp \mathbf{v}.$$

Explanation: A zero dot product means that the vectors are perpendicular.

Duality

Let

$$\mathbf{w} = \begin{bmatrix} 3 \\ -1 \end{bmatrix}.$$

Define

$$f(\mathbf{x}) = \mathbf{w}^T \mathbf{x}.$$

For

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix},$$

we get

$$f(\mathbf{x}) = 3x_1 - x_2.$$

Explanation: A vector can define a linear function using the dot product. This connection between vectors and linear functions is called **duality**.

Cross Product

The cross product is defined for vectors in $\mathbb{R}^3$.

For

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix},$$

$$\mathbf{u} \times \mathbf{v} = \begin{bmatrix} u_2 v_3 - u_3 v_2 \\ u_3 v_1 - u_1 v_3 \\ u_1 v_2 - u_2 v_1 \end{bmatrix}.$$

Example: Cross Product

Let

$$\mathbf{u} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix}.$$

Then

$$\begin{aligned} \mathbf{u} \times \mathbf{v} &= \begin{bmatrix} 2(3) - 0(1) \\ 0(0) - 1(3) \\ 1(1) - 2(0) \end{bmatrix} \\ &= \begin{bmatrix} 6 \\ -3 \\ 1 \end{bmatrix}. \end{aligned}$$

Check:

$$\mathbf{u} \cdot (\mathbf{u} \times \mathbf{v}) = 0.$$

Therefore, the resulting vector is perpendicular to $\mathbf{u}$.

Geometric Meaning of Cross Product

The magnitude is

$$\|\mathbf{u} \times \mathbf{v}\| = \|\mathbf{u}\| \|\mathbf{v}\| \sin \theta.$$

This represents the area of the parallelogram formed by $\mathbf{u}$ and $\mathbf{v}$.

Example

If

$$\|\mathbf{u}\| = 4, \quad \|\mathbf{v}\| = 5, \quad \theta = 30^\circ,$$

then

$$\|\mathbf{u} \times \mathbf{v}\| = 4(5) \sin 30^\circ$$

$$= 20 \left(\frac{1}{2} \right) = 10.$$

Therefore, the parallelogram area is 10 square units.

Cross Product and Linear Transformations

Suppose

$$\mathbf{x} \rightarrow A\mathbf{x}.$$

A linear transformation can change the area of the parallelogram formed by two vectors.

For a 3×3 matrix,

$$|\det(A)|$$

gives the volume scaling factor.

Example

Let

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Then

Cramer's Rule

Consider the system

$$ax + by = e$$

$$cx + dy = f.$$

Cramer's rule gives

$$x = \frac{\begin{vmatrix} e & b \\ f & d \end{vmatrix}}{\begin{vmatrix} a & b \\ c & d \end{vmatrix}}$$

and

$$y = \frac{\begin{vmatrix} a & e \\ c & f \end{vmatrix}}{\begin{vmatrix} a & b \\ c & d \end{vmatrix}}.$$

Cramer's Rule – New Example

Solve

$$3x + 2y = 12$$

$$x - y = 1.$$

Coefficient determinant:

$$D = \begin{vmatrix} 3 & 2 \\ 1 & -1 \end{vmatrix} = -3 - 2 = -5.$$

For x :

$$D_x = \begin{vmatrix} 12 & 2 \\ 1 & -1 \end{vmatrix} = -12 - 2 = -14.$$

Therefore,

$$x = \frac{D_x}{D} = \frac{-14}{-5} = \frac{14}{5}.$$

For y :

$$D_y = \begin{vmatrix} 3 & 12 \\ 1 & 1 \end{vmatrix} = 3 - 12 = -9.$$

Thus,

Change of Basis

A vector can be represented using different bases. Suppose

$$B = \left[\begin{array}{c|c} | & | \\ \mathbf{b}_1 & \mathbf{b}_2 \\ | & | \end{array} \right].$$

Then

$$\mathbf{x} = B[\mathbf{x}]_B.$$

Therefore,

$$[\mathbf{x}]_B = B^{-1}\mathbf{x}.$$

Explanation: The vector stays the same, but its coordinates change when the basis changes.

Change of Basis – Example

Let

$$\mathbf{b}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad \mathbf{b}_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Suppose

$$\mathbf{x} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}.$$

We need

$$\mathbf{x} = a\mathbf{b}_1 + b\mathbf{b}_2.$$

Therefore,

$$\begin{bmatrix} 4 \\ 3 \end{bmatrix} = a \begin{bmatrix} 1 \\ 1 \end{bmatrix} + b \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

This gives

$$a + b = 4, \quad a = 3.$$

Hence,

$$a = 3, \quad b = 1.$$

Therefore,

Eigenvectors and Eigenvalues

A non-zero vector $\mathbf{v}$ is an eigenvector of A if

$$A\mathbf{v} = \lambda\mathbf{v}.$$

Here:

- ▶ $\mathbf{v}$ = eigenvector
- ▶ λ = eigenvalue

The transformation changes the magnitude of $\mathbf{v}$, but not its direction.

Finding Eigenvalues

Starting with

$$A\mathbf{v} = \lambda\mathbf{v},$$

we obtain

$$(A - \lambda I)\mathbf{v} = 0.$$

For a non-zero vector $\mathbf{v}$,

$$\det(A - \lambda I) = 0.$$

This is called the characteristic equation.

Eigenvalue Example

Consider

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

Then

$$A - \lambda I = \begin{bmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{bmatrix}.$$

Therefore,

$$\det(A - \lambda I) = (2 - \lambda)^2 - 1.$$

Set equal to zero:

$$(2 - \lambda)^2 - 1 = 0.$$

Thus,

$$2 - \lambda = \pm 1.$$

Finding Eigenvectors

For

$$\lambda_1 = 3,$$

solve

$$(A - 3I)\mathbf{v} = 0.$$

We get

$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Therefore,

$$x = y.$$

Choose $x = 1$:

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

For $\lambda_2 = 1$,

$$x = -y.$$

Therefore,

$$\mathbf{v}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Eigenvectors in PCA

Principal Component Analysis (PCA) uses eigenvectors of the covariance matrix.
If

$$S\mathbf{v} = \lambda\mathbf{v},$$

then:

- ▶ Eigenvectors give principal directions.
- ▶ Eigenvalues indicate the amount of variance.

Simple idea:

A large eigenvalue indicates a direction in which the data has large variation.

Vector Spaces

A vector space is a set of objects together with vector addition and scalar multiplication satisfying certain rules.

Examples include:

$$\mathbb{R}^2, \quad \mathbb{R}^3, \quad \mathbb{R}^n.$$

Example

The set

$$\mathbb{R}^2 = \left\{ \begin{bmatrix} x \\ y \end{bmatrix} : x, y \in \mathbb{R} \right\}$$

is a vector space.

Subspaces

A subset W of a vector space V is a subspace if:

1. $\mathbf{0} \in W$
2. $\mathbf{u}, \mathbf{v} \in W \Rightarrow \mathbf{u} + \mathbf{v} \in W$
3. $\mathbf{u} \in W \Rightarrow c\mathbf{u} \in W$

Example

Consider

$$W = \left\{ \begin{bmatrix} x \\ 2x \end{bmatrix} : x \in \mathbb{R} \right\}.$$

This is a line through the origin.

Therefore, W is a subspace of $\mathbb{R}^2$.

Dimension and Basis

The dimension of a vector space is the number of vectors in any basis.
For example,

$$\mathbb{R}^2$$

has the standard basis

$$\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}.$$

Therefore,

$$\dim(\mathbb{R}^2) = 2.$$

Similarly,

$$\dim(\mathbb{R}^3) = 3.$$

Rank and Nullity

For

$$A \in \mathbb{R}^{m \times n},$$

$$\text{rank}(A) = \dim(\text{Col}(A))$$

and

$$\text{nullity}(A) = \dim(\text{Null}(A)).$$

The Rank-Nullity Theorem is

$$\boxed{\text{rank}(A) + \text{nullity}(A) = n}$$

where n is the number of columns.

Rank-Nullity Example

Consider

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}.$$

The second row is twice the first row.

Therefore,

$$\text{rank}(A) = 1.$$

Since A has 3 columns,

$$\text{rank}(A) + \text{nullity}(A) = 3.$$

Hence,

$$1 + \text{nullity}(A) = 3.$$

Therefore,

$$\boxed{\text{nullity}(A) = 2}.$$

Linear Algebra in AI/ML

Linear algebra appears throughout machine learning.

Data $\rightarrow$ Vectors/Matrices $\rightarrow$ Transformations $\rightarrow$ Model $\rightarrow$ Prediction

Examples:

- ▶ Vectors $\rightarrow$ feature representation
- ▶ Matrices $\rightarrow$ datasets and parameters
- ▶ Matrix multiplication $\rightarrow$ neural networks
- ▶ Dot products $\rightarrow$ similarity and prediction
- ▶ Eigenvectors $\rightarrow$ PCA
- ▶ Nonsquare matrices $\rightarrow$ dimensionality reduction

AI/ML Example: Linear Prediction

Suppose

$$\mathbf{x} = \begin{bmatrix} 2 \\ 4 \\ 3 \end{bmatrix}$$

is an input vector and

$$\mathbf{w} = \begin{bmatrix} 0.5 \\ 1 \\ 2 \end{bmatrix}.$$

Let

$$b = 2.$$

The prediction is

$$\hat{y} = \mathbf{w}^T \mathbf{x} + b.$$

Therefore

Important Formulas – Vectors

Vector Addition

$$\mathbf{u} + \mathbf{v} = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \end{bmatrix}$$

Scalar Multiplication

$$c\mathbf{v} = \begin{bmatrix} cv_1 \\ cv_2 \end{bmatrix}$$

Magnitude

$$\|\mathbf{v}\| = \sqrt{\mathbf{v}^T \mathbf{v}}$$

Linear Combination

$$\sum_{i=1}^k a_i \mathbf{v}_i.$$

Important Formulas – Matrices

Transformation

$$\mathbf{y} = A\mathbf{x}$$

Determinant

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$$

Inverse

$$AA^{-1} = I$$

Null Space

$$\text{Null}(A) = \{\mathbf{x} : A\mathbf{x} = 0\}.$$

Important Formulas – Products

Dot Product

$$\mathbf{u}^T \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta.$$

Cross Product

$$\|\mathbf{u} \times \mathbf{v}\| = \|\mathbf{u}\| \|\mathbf{v}\| \sin \theta.$$

Eigenvalue Equation

$$A\mathbf{v} = \lambda\mathbf{v}.$$

Characteristic Equation

$$\det(A - \lambda I) = 0.$$

Important Formula – Change of Basis

If B is the basis matrix,

$$\mathbf{x} = B[\mathbf{x}]_B.$$

Therefore,

$$[\mathbf{x}]_B = B^{-1}\mathbf{x}$$

provided that B is invertible.

Interpretation:

The vector $\mathbf{x}$ does not change.

Only its coordinate representation changes.

Final Summary

- ▶ Vectors represent quantities and data points.
- ▶ Coordinate systems provide representations of vectors.
- ▶ Vector addition combines vectors component-wise.
- ▶ Scalar multiplication changes magnitude and direction.
- ▶ Linear combinations construct new vectors.
- ▶ Span describes all obtainable linear combinations.
- ▶ A basis is linearly independent and spanning.
- ▶ Matrices represent linear transformations.
- ▶ Matrix multiplication represents composition.
- ▶ Determinants describe scaling and invertibility.

Final Summary – Continued

- ▶ Inverse matrices reverse invertible transformations.
- ▶ Column space describes possible outputs.
- ▶ Null space describes inputs mapped to zero.
- ▶ Nonsquare matrices map between different dimensions.
- ▶ Dot products measure alignment.
- ▶ Cross products produce perpendicular vectors in $\mathbb{R}^3$.
- ▶ Cramer's rule solves certain linear systems.
- ▶ Change of basis gives alternative coordinate systems.
- ▶ Eigenvectors identify invariant directions.
- ▶ Eigenvalues describe scaling along those directions.
- ▶ Vector spaces provide the mathematical framework for linear algebra.

Practice Problems

Problem 1

Given

$$\mathbf{u} = \begin{bmatrix} 3 \\ -2 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 1 \\ 4 \end{bmatrix},$$

find:

1. $\mathbf{u} + \mathbf{v}$
2. $2\mathbf{u} - \mathbf{v}$
3. $\|\mathbf{u}\|$

Problem 2

Determine whether

$$\begin{bmatrix} 1 \\ 3 \end{bmatrix}, \quad \begin{bmatrix} 2 \\ 6 \end{bmatrix}$$

are linearly independent.

Practice Problems – Continued

Problem 3

For

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix},$$

calculate:

1. $\det(A)$
2. A^{-1}

Problem 4

Find the eigenvalues and eigenvectors of

$$A = \begin{bmatrix} 5 & 0 \\ 0 & 2 \end{bmatrix}.$$

Conceptual Questions

1. What is the difference between a scalar and a vector?
2. What does the span of vectors represent?
3. What conditions are required for a basis?
4. Why does matrix multiplication represent composition?
5. What does the determinant tell us?
6. What is the difference between column space and null space?
7. Why can nonsquare matrices represent transformations?
8. What is the geometric meaning of the dot product?
9. What is an eigenvector?
10. What does an eigenvalue represent?

AI/ML Applications of Linear Algebra

Concept	Application
Vectors	Feature representation
Matrices	Dataset representation
Linear combinations	Linear models
Matrix multiplication	Neural networks
Dot product	Similarity and prediction
Nonsquare matrices	Dimension mapping
Determinant	Transformation geometry
Null space	Redundancy analysis
Eigenvectors	PCA
Basis	Feature representation

Why Linear Algebra Matters in AI

Almost every modern machine-learning model uses linear algebra.

Data $\rightarrow$ Vectors $\rightarrow$ Matrices $\rightarrow$ Transformations $\rightarrow$ Learning

- ▶ Neural networks use matrix multiplication.
- ▶ Regression uses vectors and matrices.
- ▶ PCA uses eigenvectors and covariance matrices.
- ▶ Embeddings are represented as high-dimensional vectors.
- ▶ Similarity search uses dot products.
- ▶ Computer vision uses matrix transformations.

Thank You